

SciFig-Evaluation: A Multilingual Benchmark and Behavioural Framework for Evaluating Multimodal Language Models on Scientific Figures

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Declaration

No portion of the work contained in this document has been submitted in support of an application for a degree or qualification of this or any other university or other institution of learning. All verbatim extracts have been distinguished by quotation marks, and all sources of information have been specifically acknowledged.

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Abstract

Multimodal large language models are increasingly asked to read scientific figures, yet existing benchmarks largely measure whether a model can extract a value from a chart, not whether it is a faithful visual reader. This thesis introduces SCIFIG-EVAL, a multilingual benchmark of 1,005 figures drawn from 158 arXiv papers and paired with 1,411 expert annotations in English, Bulgarian, German and Chinese, together with a three-axis behavioural framework — Admittance, Resistance and Inductance — that separates honesty in the absence of evidence, faithfulness under misleading context, and sound inductive reasoning. Using an MQM-style multi-judge evaluation pipeline, the framework is exercised on eleven contemporary models spanning five families via five adversarial probe types. The resulting behavioural profile reveals systematic, near-orthogonal failure modes that conventional accuracy scores obscure.

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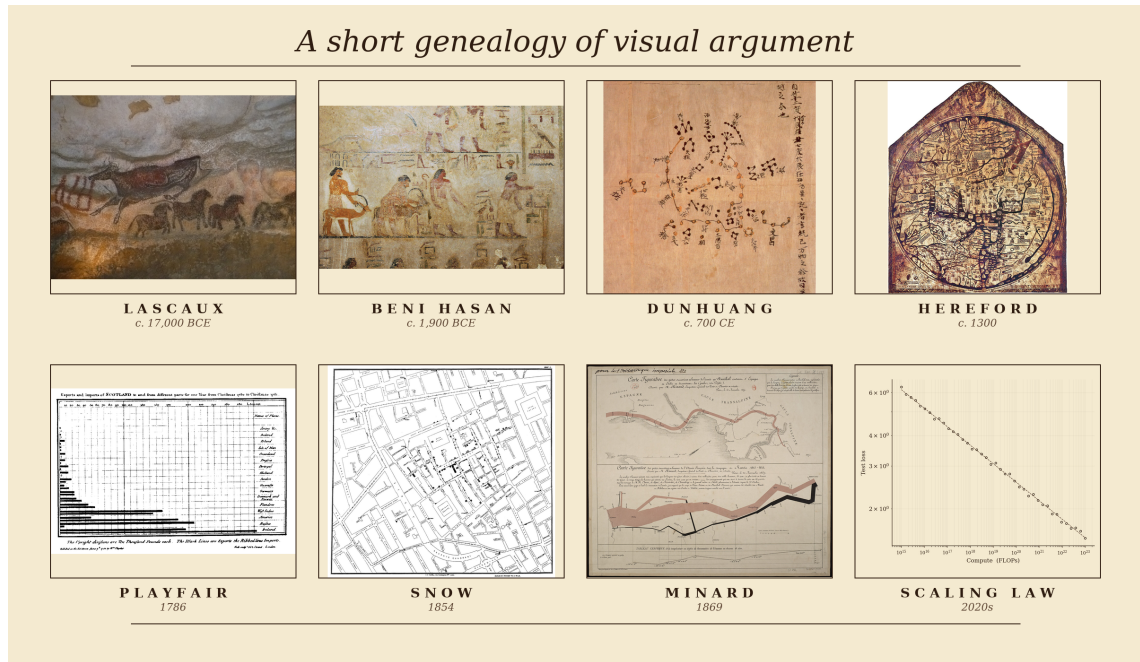
Chapter 1

Introduction

1.1 *Homo sapiens*: A Species of Visual Thinkers

The record of a species willing to reason pictorially begins no later than the Upper Palaeolithic. Some time between seventeen and fifteen millennia before the present, an inhabitant of what is now the Dordogne crushed haematite between stones, lit a pine torch, and rendered an aurochs on Lascaux limestone at approximately life size (Aujoulat, 2005). These visual arguments have continued to evolve. Twelfth-dynasty Egyptian painters at Beni Hasan committed a caravan of Semitic travellers, traditionally associated with the Joseph narrative, to a tomb wall around 1900 BCE (Kamrin, 2009); seventh-century Chinese astronomers committed over fourteen hundred stars to silk at Dunhuang (Bonnet-Bidaud et al., 2009); medieval cartographers compressed theology, geography and chronology onto the Hereford *mappa mundi* (Harvey, 1996). In 1786 William Playfair invented the bar chart in order to argue, graphically and in print, about Britain's trade balances (Playfair, 1786). John Snow's 1854 cholera map of Soho laid down the method of modern epidemiology (Johnson, 2006), and Charles Joseph Minard's 1869 flow map of Napoleon's 1812 Russian campaign (Friendly, 2002) is what Tufte (2001) has called, without serious contest, the densest argument ever made on paper (Figure 1.1a). When a human being intends a claim of any real quantitative weight, the first move is almost invariably to draw.

In contemporary practice, figures have not only evolved but have become significantly embedded in daily operations. Businesses read through dashboards of candlestick charts and cohort curves (Nison, 2001) that senior management consults in direct preference to prose; engineers coordinate the hundreds of thousands of components that constitute a ship, a hospital or an aircraft through hierarchies of blueprints read without ambiguity by people who may never speak to the designer (Ferguson, 1992); and even life-and-death decisions are made by reading a heartbeat off the grid of an electrocardiogram, or tissue contrast off a radiograph or perfusion map (Goldberger et al., 2017). Music (Treitler, 1984), satellite navigation (Kaplan and Hegarty, 2017), chemistry (Hoffmann, 1995) and network science (Newman, 2010) sustain the same diagrammatic primacy, and every competent empirical paper in the natural and social sciences eventually commits its argument to a figure (Figure 1.1b). **The figure has become the lingua franca of modern knowledge work**, and in many domains the argumentative weight it now carries is strictly greater than the weight carried by the accompanying prose.



The figure as the lingua franca of modern knowledge work

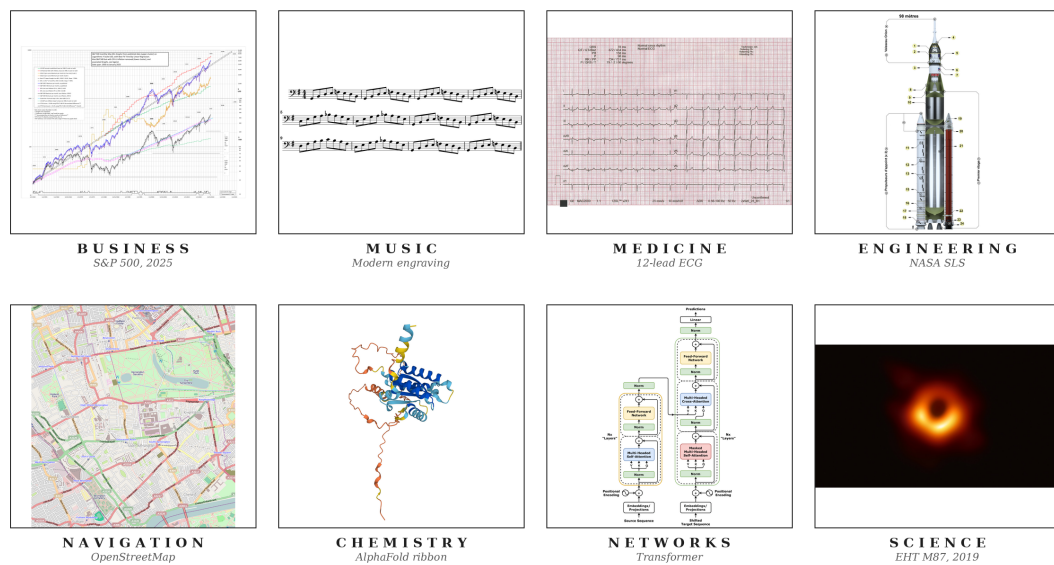


Figure 1.1: (a) A short genealogy of visual argument, from Lascaux and Egyptian wall painting through the medieval *mappa mundi*, Playfair’s 1786 bar chart and Snow’s 1854 cholera map to modern neural scaling curves. (b) The figure as lingua franca of modern knowledge work across business, music, medicine, engineering, navigation, chemistry, networks and the natural sciences. *Sources.* (a) Lascaux aurochs, Beni Hasan caravan, Dunhuang star chart, Hereford *mappa mundi*, Playfair 1786 bar chart, Snow 1854 cholera map and Minard 1869 flow map of Napoleon’s Russian campaign, all public domain (via Wikimedia Commons, the Metropolitan Museum of Art, the British Library, Hereford Cathedral, the Internet Archive and the Bibliothèque nationale de France respectively); scaling curves adapted from Kaplan et al. 2020. (b) A candlestick chart, score excerpt, ECG trace, blueprint, nautical chart, molecular diagram, network graph and scientific plot, from public-domain or openly-licensed repositories (Wikimedia Commons, NOAA, PhysioNet) or rendered by the author.

1.2 Can *Intelligentia Machinae* (Machine Intelligence) Comprehend Visuals Similarly?

Recent progress in artificial intelligence has been uncommonly rapid, and the current generation of large models has brought within plausible reach an older ambition of the field: a machine that can collaborate with a human being to achieve a shared task. A necessary condition of that collaboration is that the two parties reason on similar planes and navigate near-similar cognitive paradigms. Exact cognitive parity is not required; only a proximity close enough to sustain useful work in concert with a human collaborator. Since the human, as the previous section establishes, is a visual thinker, that proximity obliges the machine to comprehend the visual: to read, interpret and reason from the figure with fidelity comparable to its reading of text.

Systems at the current frontier, including Anthropic’s Claude Opus 4.6 and 4.7 (Anthropic, 2026a,b), OpenAI’s GPT-5 series (OpenAI, 2025a) and Google DeepMind’s Gemini 3 Pro (Google DeepMind, 2026a), are now deployed as infrastructure rather than laboratory curiosities. They are shipped into enterprise software through the major cloud providers, and they are measurably altering how code is written and how commercial operations are conducted. The salient question is therefore no longer whether such systems are capable, but whether that capability extends, in the way the human partner’s does, to the visual.

1.3 Motivation for Focusing on Scientific Research Figures

Research is among the first domains in which collaborative intelligence is being tested at scale: between late 2024 and early 2026 the three largest model developers each shipped autonomous research agents (Google, 2024; OpenAI, 2025c; Anthropic, 2025), and scientific deployments such as the Biomni biomedical platform now report research-cycle compressions of an order of magnitude or more (Anthropic, 2026d). Yet scientific knowledge is not monolingual; a substantial share of the record is produced in Chinese, German, Bulgarian and other languages of research, and the cost of excluding that share, in citation, uptake and participation, has been documented at length (Amano et al., 2023). Chart- and figure-understanding benchmarks, however, remain predominantly English: recent multilingual efforts such as PolyChartQA (Li et al., 2025), PM4Bench (Gao et al., 2025) and IndicVisionBench (Sharma et al., 2025) each report substantial cross-lingual degradation on tasks the same models solve comfortably in English. The reach of a multimodal system, considered as a research collaborator, therefore scales with the breadth of languages over which its visual reading remains reliable, and the scientific figure, considered across languages, is the natural object at which to test whether the current frontier reads visually or merely appears to. What that reading must withstand, beyond correctness on the clean image, is illustrated in Figure 1.2.

The failure is not a miscount or a mislabelled axis; it is a confident substitution of a plausible category for evidence the model cannot actually see. Honest reading, not just accurate reading, is therefore the object of interest, and the research questions below are framed accordingly: they ask how faithfully today’s frontier models describe scientific figures across languages; how their reading survives visual degradation and adversarial context; and whether, when the evidence is not there, they acknowledge it rather than fill it in.

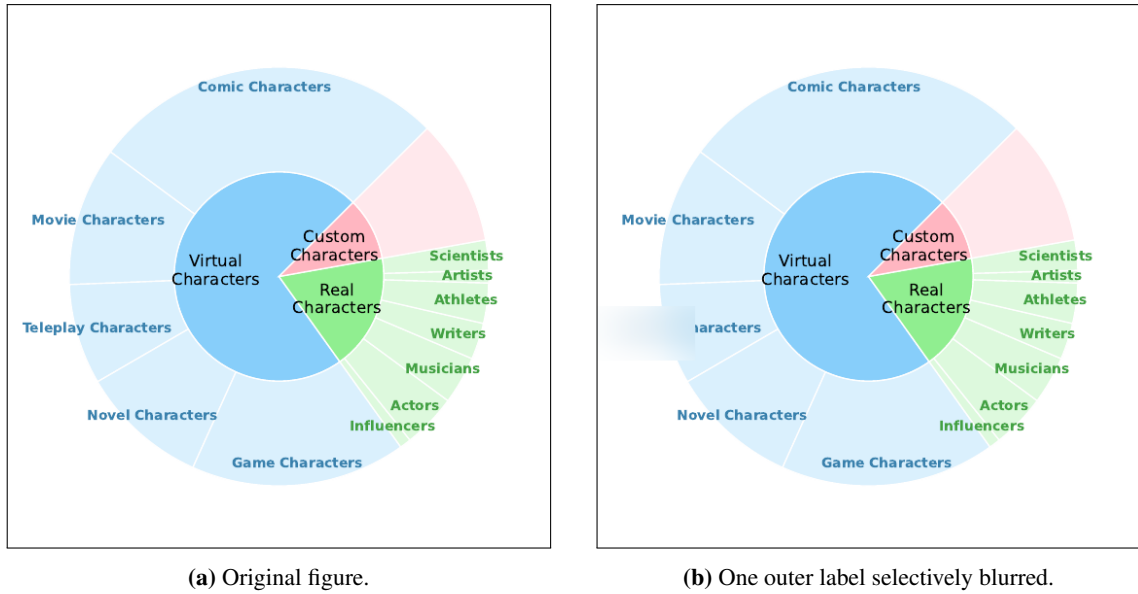


Figure 1.2: A sunburst taken from an arXiv paper in the SCIFIG-EVAL corpus. In (a), the outer-ring category between *Movie Characters* and *Novel Characters* is legibly *Teleplay Characters*; in (b), that single label has been blurred. Asked which category sits between the two, GPT-5.2 answers “*Anime Characters*,” with no admission that the relevant region is unreadable. Source image reproduced from arXiv:2505.23923v1 under its open-access licence; blurred variant generated by the SCIFIG-EVAL pipeline.

1.4 Research Questions

The central objective of this thesis is to characterise, across models, languages and adversarial conditions, how faithfully today’s frontier multimodal models actually read scientific figures, and to build an evaluation framework that surfaces the failure modes which conventional accuracy benchmarks obscure. Concretely, the investigation is organised around the following five research questions, each pairing a capacity with the conditions under which that capacity is tested:

RQ1 (Description accuracy across languages). To what extent do current vision–language models produce accurate and complete descriptions of scientific figures across typologically diverse languages?

RQ2 (Robustness to visual degradation). How does visual degradation, including noise, compression, aspect distortion and contextual embedding within paper pages, affect the accuracy (whether what is stated is correct), completeness (whether what is visible is reported) and clarity (whether what is reported is expressed unambiguously) of VLM-generated figure descriptions?

RQ3 (Comprehension beyond description). Beyond surface-level description, can VLMs demonstrate genuine comprehension of scientific visualisations, from answering quantitative questions to performing non-trivial inductive reasoning that requires distinguishing analytical insight from surface-level pattern matching?

RQ4 (Adversarial robustness). How robust are VLM outputs when subjected to adversarial manipulations, including misleading captions, contradictory premises and psychologically-informed sycophancy probes?

RQ5 (Calibrated epistemic honesty). When confronted with incomplete or unresolvable visual information, do VLMs exhibit calibrated epistemic honesty, acknowledging uncertainty rather than confabulating plausible but ungrounded responses?

1.5 Thesis Structure

The remainder of the thesis is organised as follows. Chapter 2 (Background) provides the technical and conceptual foundation: how frontier multimodal models convert figures into internal representations, how scientific figures carry argument rather than merely depict it, and the failure modes and evaluation traditions on which the remainder draws. Chapter 3 (Literature Survey) situates the work against chart understanding, multimodal hallucination, multilingual evaluation and MQM-style scoring. Chapter 4 (Design) sets out the SciFig corpus, the A–R–L framework and the five adversarial probe families. Chapter 5 (Implementation) describes the evaluation pipeline, the model and judge harness and the reproducibility controls. Chapter 6 (Evaluation) reports the main quantitative results across the 11 evaluated models and the four languages, together with ablations on judge ensembles and probe families. Chapter 7 (Discussion) interprets the findings, connects them to the A–R–L framework and acknowledges the limitations of the study. Chapter 8 (Conclusion) summarises the contributions, revisits the research questions and outlines directions for future work.

Chapter 2

Background

This chapter lays the technical and conceptual ground for the behavioural evaluation in Chapters 4–6. Section 2.1 describes how today’s frontier multimodal models turn an image into the internal representation on which every downstream capability depends. The remaining sections, in draft, treat the scientific figure as an object of visual argument; distinguish description from reasoning as cognitive acts; survey the failure modes that shape this thesis (hallucination, sycophancy and mis-calibration); introduce MQM-style open-ended evaluation and the LLM-as-judge methodology; and close on the particular complications of reading text embedded in a figure whose language is not English.

2.1 How do today’s Large Language Models see?

To understand the visual capability of today’s Large Language Models, a useful starting point is to grasp how they see. Every family evaluated in this thesis shares the common skeleton sketched in Figure 2.1: a raster image is partitioned into patches by a vision encoder, the resulting embeddings are retuned by a connector into the token space of a language backbone, and the backbone consumes those projected vision tokens as a prefix alongside the text prompt (Liu et al., 2023; Alayrac et al., 2022; Li et al., 2023). From the backbone’s point of view, the image is, by the time it arrives, a sequence of soft tokens drawn from the same vector space as text.

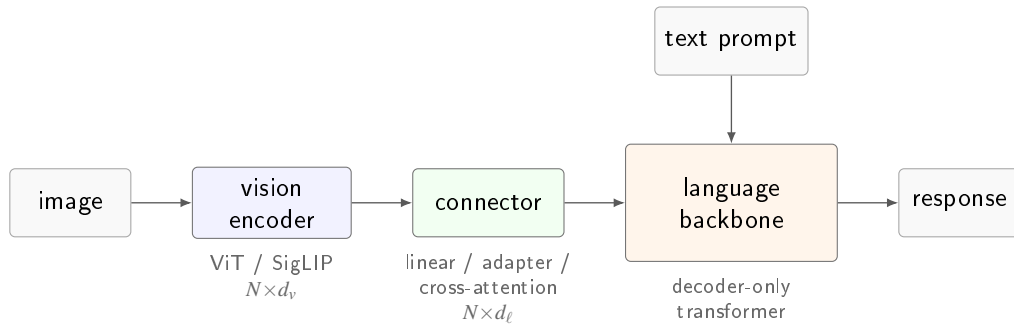


Figure 2.1: The canonical vision-language skeleton shared by every family evaluated in this thesis. A vision encoder, typically a Vision Transformer (Dosovitskiy et al., 2021) trained with a CLIP (Radford et al., 2021) or SigLIP (Zhai et al., 2023) contrastive objective, maps the image to N patch embeddings of width d_v ; a connector (a linear projection, a lightweight adapter or a cross-attention layer) retunes them to the backbone width d_l (Liu et al., 2023; Li et al., 2023); the decoder-only language model consumes them as a prefix alongside the text prompt. Families differ in how each stage is realised, not in the skeleton itself.

The encoder. The encoder is, across every family evaluated here, a Vision Transformer (Dosovitskiy et al., 2021), trained either with the contrastive image-text objective of CLIP (Radford et al., 2021) or with its sigmoid variant SigLIP (Zhai et al., 2023). SigLIP replaces the softmax contrastive loss with an independent binary classification on every image-text pair, removing the quadratic coupling across a batch, and has become the default front end for Gemma 3 (Gemma Team, Google DeepMind, 2025) and Phi-4 multimodal (Microsoft Research, 2025). Qwen3-VL (Qwen Team, Alibaba, 2025), building on Qwen2-VL (Bai et al., 2024), departs from this recipe with a ViT trained from scratch and a two-dimensional rotary position encoding (2D-RoPE) that lets the encoder ingest images of arbitrary resolution and aspect ratio without rescaling. Whichever objective is used, the encoder emits a sequence of patch embeddings whose length scales with the image area and whose width is fixed by the encoder.

The connector. The connector bridges the encoder’s embedding space to the backbone’s. Three designs predominate. A *linear projection* (or two-layer MLP), as in LLaVA (Liu et al., 2023), is the minimal option and preserves every patch token one-for-one. A *cross-attention adapter* inserts a small set of learnable queries that attend to the encoder output and deliver a fixed-length summary to the backbone, first popularised by Flamingo (Alayrac et al., 2022) and refined by the Q-Former in BLIP-2 (Li et al., 2023). A *low-rank adapter* grafts LoRA modules (Hu et al., 2022) into the backbone itself and activates them only when a vision input is present, as in Phi-4 multimodal’s mixture-of-LoRAs design (Microsoft Research, 2025). The choice is far from cosmetic: the connector dictates how many image tokens reach the backbone, whether those tokens are summarised or preserved verbatim, and how much of the backbone is fine-tuned on multimodal data.

The fusion regime. Families further differ in *when* the vision stream meets the language stream. The simplest regime, *adapter fusion*, grafts a pretrained vision encoder onto a frozen (or lightly tuned) language backbone through a late connector; Gemma 3 and the original LLaVA are representative. *Early-fusion* models interleave image and text tokens from the first pre-training step, giving the backbone a genuinely bimodal token distribution; Llama 4 (Meta AI, 2025) and Chameleon (Chameleon Team, 2024) are the public exemplars. *Native multimodal* training, claimed by Qwen3-VL and by the closed frontier families, scales this further: the visual stack is co-trained with the backbone from scratch on a corpus that was never unimodal. The three regimes trade data efficiency for representational depth, and, as later sections will argue, they translate into different failure modes at read-time.

Within this common skeleton, the seven families evaluated here diverge along four axes that matter empirically: the choice of vision encoder; the input-resolution strategy (a fixed tile with Pan-and-Scan reassembly (Gemma Team, Google DeepMind, 2025), dynamic tiling that preserves aspect ratio (Bai et al., 2024), or encoder-native arbitrary resolution); the per-image token budget delivered to the backbone; and the fusion regime. Table 2.1 collects what each family publicly discloses along these axes.

The table is as informative for what it omits as for what it contains. The four open-weight families each publish a technical report that documents the vision stack end to end; the three closed frontier families publish system cards (Anthropic, 2026c; OpenAI, 2025b; Google DeepMind, 2026b) oriented toward capability and safety, and leave the encoder, the resolution policy, the token

Table 2.1: Publicly documented vision-language architecture across the seven families evaluated in this thesis. “–” denotes a detail that is not disclosed in the primary source. The four open-weight families (top block) publish detailed technical reports; the three closed frontier families (bottom block) publish system cards that address capability and safety but leave the vision stack undisclosed. *Sources:* Gemma 3 (Gemma Team, Google DeepMind, 2025); Qwen3-VL (Qwen Team, Alibaba, 2025), with architectural continuity from Qwen2-VL (Bai et al., 2024); Llama 4 (Meta AI, 2025); Phi-4 multimodal (Microsoft Research, 2025); Claude Opus 4.x (Anthropic, 2026c); GPT-5 / 5.2 (OpenAI, 2025b); Gemini 3 Pro (Google DeepMind, 2026b).

Family	Vision encoder	Resolution / tiling	Image tokens	Fusion regime
Gemma 3	frozen SigLIP-400M	896×896 + Pan&Scan	256 (linear proj.)	adapter
Qwen3-VL	custom ViT + 2D-RoPE	dynamic, any aspect	variable (MRoPE)	native
Llama 4	MetaCLIP-derived ViT	–	– (early-fusion)	native early-fusion
Phi-4 mm	CLIP variant + LoRA	–	–	Mixture-of-LoRAs
Claude Opus 4.x	–	≤ 2576 px long edge	–	–
GPT-5 / 5.2	–	–	–	native (claimed)
Gemini 3 Pro	–	up to 900 images / prompt	–	native (sparse MoE)

budget and the connector architecture undisclosed. The asymmetry is structural: the models most widely deployed in real research settings are the models whose visual reading is architecturally least inspectable. An evaluation that required internal access would, at the frontier, have almost nothing to evaluate.

This thesis treats that constraint as a design requirement. Chapter 4 develops a three-axis behavioural framework, Admittance, Resistance and Inductance, that characterises a model’s visual reading from the outside, treating the model as a circuit whose internal structure need not be known in order to diagnose how it fails. The architectural variations in Table 2.1 return at the end of this chapter, where we ask how specific design choices, from aggressive image-token compression through adapter-only vision heads to fixed-tile resolution, translate into predictable failure modes at read-time.

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